AI evidence pack — certified anytime perception

Prepared for Prof. Houben — 2026-10-23. Every number below is read from the saved result files at build time.

0. What is AI here, and what is not

The system uses learned components in four places and deliberately does not use learning in two others. Stating the split first, because it is the substance of the contribution rather than a caveat.

Component	Mechanism	Learned how
Multi-exit perception model	3-exit MobileNetV1 CNN	**Gradient descent**, 100 epochs, Keras
Baseline control model	Same backbone, single exit	**Gradient descent**
Conformal certification (tau)	Split-conformal quantile over calibration split	Not learned — finite-sample quantile, distribution-free
Learned exit policy	Isotonic (PAVA) risk curves $risk_k(c)$	Monotone curve fit on calibration only
TTCA / ACI adaptation	Online alpha update from observed errors	Online update rule, no training
Joint exit + security-level scheduler	Policy over measured costs	Trace-driven simulator
Leakage classifiers (adversary)	RandomForest / MLP on power envelopes	Gradient / tree fit

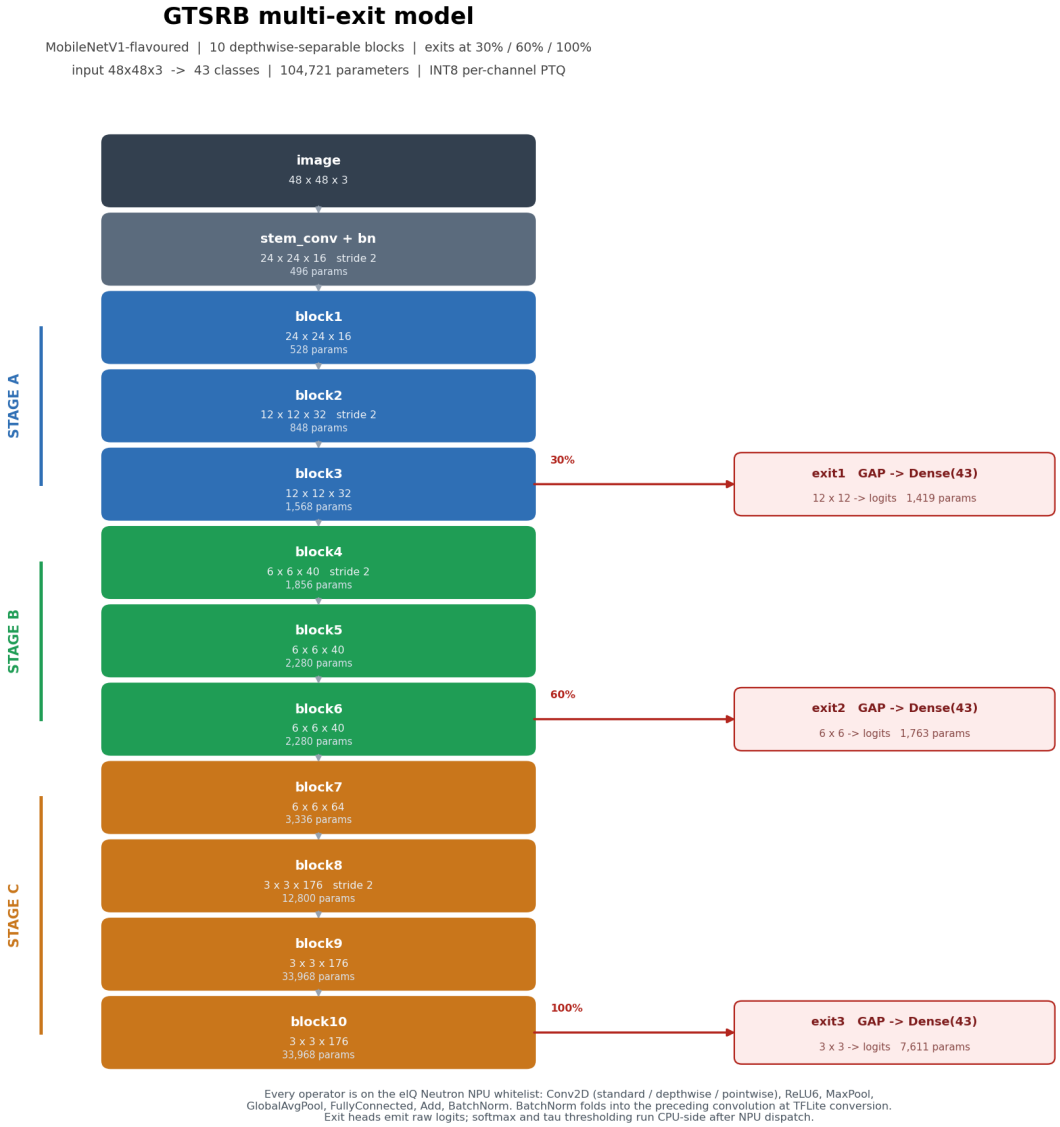
1. The perception model

A three-exit MobileNetV1-flavoured backbone, built so that every operator is on the eIQ Neutron NPU whitelist. Exits sit at **exactly** 30% / 60% / 100% of block depth (blocks 3, 6, 10 of 10), so the anytime tiers are structural rather than approximate.

Property	Value
Input / classes	48 x 48 x 3, INT8 → 43 GTSRB classes
Depth	10 depthwise-separable blocks, 3 stages
Stage widths	stem 16 → A 32 → B 40 → C 176
Parameters	**104,721** (verified against the checkpoint's weight count)
Exit head cost	exit1 1,419 · exit2 1,763 · exit3 7,611 params
INT8 device size	segA 16,736 B · segB 21,168 B · segC 125,896 B

The unequal stage widths are a deliberate design choice, not an accident of tuning. Stage B was narrowed from 64 to 40 channels and stage C widened from 128 to 176 because at 64 channels exits 2 and 3 converged to within ~1 percentage point by epoch 15 — the exit-collapse failure mode. Without a capacity handicap on the early exits, GTSRB is easy enough that the anytime property becomes vacuous.

Architecture



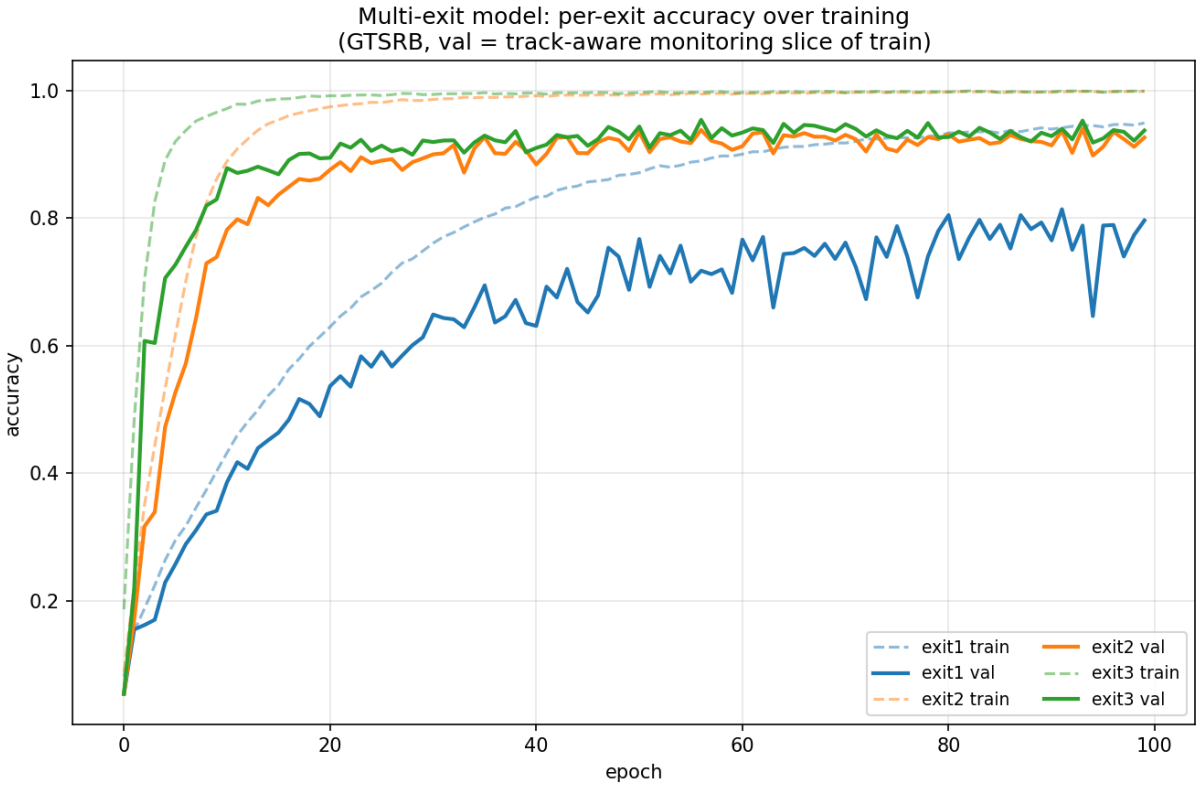
One box per depthwise-separable block (depthwise conv + BN + pointwise conv + BN). Exit heads take a global average pool then a single Dense(43); they emit raw logits, so softmax and tau thresholding happen CPU-side after NPU dispatch.

2. Training

Setting	Value
Seed / batch / epochs / lr	20260728 / 128 / 100 / 0.001
Loss	BranchyNet joint loss, $L = \Sigma_k w_k \cdot CE(exit_k)$, $w_k = 1.0$
Validation	track-aware 10% holdout taken from the training split only
Checkpoint rule	MaxOfMin — maximises min(val exit1, exit2, exit3), protects the weakest exit
Selected epoch	95 (min-of-exits = 0.733, set by exit1)

The validation split is track-aware on purpose. GTSRB frames are named `<trackID>_<frameID>.ppm` with roughly 30 near-duplicate frames per physical sign, so a random split leaks the same sign into both sides and inflates validation accuracy. Whole tracks are held out instead.

Best validation accuracy: **exit1 0.733 · exit2 0.923 · exit3 0.931**. The single-exit baseline control reaches **0.945**, so joint multi-exit training costs the final classifier **~1.4 points**. That is the measured price of the anytime property, and the number to quote when asked whether early exits are free.



Per-exit accuracy over training.



Per-exit loss over training.

3. Conformal certification — the safety guarantee

This is the part that turns an early-exit network into a system with a stated risk bound. For each (exit, alpha) pair, split-conformal gives the threshold tau such that the empirical safety-critical miss rate among accepted samples is bounded by alpha, using the finite-sample correction $(n \cdot \hat{R} + 1)/(n + 1) \leq \text{alpha}$. tau is computed on the calibration split only; the test split is touched once.

exit	alpha	tau	accepted (test)	achieved SC miss	verdict
exit1	0.01	1.0	0	n/a	fail
exit1	0.05	0.858	2,928	0.0447	**PASS**
exit1	0.1	0.723	4,303	0.0904	**PASS**
exit2	0.01	0.995	5,795	0.0112	fail
exit2	0.05	0.815	7,530	0.0517	fail
exit2	0.1	0.402	8,399	0.0985	**PASS**
exit3	0.01	1.0	2,616	0.0027	**PASS**
exit3	0.05	0.912	7,730	0.0450	**PASS**
exit3	0.1	0.0	8,460	0.0887	**PASS**

6 of 9 tiers certify under the frozen protocol. The three that do not are reported as failures rather than tuned away:

Tier	Why it fails
exit1 @ 0.01	not certifiable at calibration — zero accepted samples survive the bound
exit2 @ 0.01	achieved 0.0112 > 0.01 on test
exit2 @ 0.05	achieved 0.0517 > 0.05 on test

The exit2 @ 0.05 failure is the interesting one. It passed calibration and then missed the bound on test — exactly the calibration-to-deployment shift that a certificate is supposed to expose. Under the earlier, contaminated protocol this tier had appeared to pass. It is reported here as a failure.

4. Learned exit policy

Instead of a fixed tau per exit, the policy learns the risk curve $\text{risk}_k(c) = P(\text{safety-critical AND wrong} \mid \text{confidence}_k = c)$ as an isotonic (PAVA, decreasing) fit on the calibration split, then takes the shallowest exit whose learned risk is under the target alpha.

This policy was evaluated on the test set, which had already been touched once. Rather than re-use the same alpha, the second look is **Bonferroni-corrected to alpha/2 = 0.025**:

exit	SC accepted	SC miss rate	corrected target	verdict
exit1	1,567	**1.47%**	≤ 2.5%	**certifies**
exit2	5,342	**3.61%**	≤ 2.5%	does not certify

So the learned policy adds a certifying operating point at exit1 (1.47% against a 2.5% budget) and an exit2 point that does **not** certify. Reporting the exit2 result as uncertified is the honest reading; the uncorrected 3.99% figure it was first quoted with was a double-touch artifact.

5. Adaptive conformal (TTCA / ACI) under distribution shift

A static conformal threshold assumes exchangeability. When the input distribution shifts, coverage collapses. ACI restores it by updating alpha online from observed errors: $\text{alpha}_{t+1} = \text{alpha}_t + \text{gamma}(\text{alpha}_{\text{target}} - \text{I}(Y_t \notin C_t))$.

segment	static conformal	ACI (TTCA)
clean, frames 0–299	0.958	0.958
shift, 300–699	**0.486**	**0.939**
post-shift, 700–999	0.439	**0.949**
ALL	0.596	**0.947**

Static coverage collapses from 95.8% to 48.6% under shift; ACI holds 93.9–94.9%. The cost is honest and quantified: **+3.3% compute** (0.80 → 0.83 ms/frame) and the prediction set inflates from 4.78 to 26.8 of 43 classes. ACI is also **reactive, not predictive**: coverage is only 86% in the first 50 frames after the shift and reaches 95% after about 100 frames. That lag is a property of the method and is reported rather than smoothed over.

6. A negative result we acted on: the LTC is a NO-GO

A continuous-time model (LTC) was proposed as a way to get more accuracy per unit of compute. Before building it, the existing exit frontier was audited:

exit	depth	latency	accuracy
exit1	30%	0.80 ms	73.3%
exit2	60%	1.24 ms	92.2%
exit3	100%	2.34 ms	93.1%

exit2 already captures 99.0% of exit3's accuracy at 53% of the cost. Only +0.9 percentage points remain in the entire exit2 → exit3 interval. For a continuous model to win on efficiency it would have to reach exit2's accuracy at exit1's cost — a 20-point jump at 0.80 ms. The frontier saturates, so the LTC was rejected **before** spending the training budget on it. Verdict: **NO-GO**.

7. Learned joint scheduling: perception and crypto share one deadline

The model's exits and the cryptographic handshakes compete for the same frame budget. A trace-driven scheduler over measured costs decides, per frame, which exit to take and, per handshake, which ML-KEM security level to use.

input	value
exit latency	0.80 / 1.24 / 2.34 ms
ML-KEM-512 / 768 / 1024 handshake	72.2 / 116.0 / 174.9 ms
energy per handshake	4.20 / 6.63 / 9.76 mJ
frame deadline (30 fps)	33 ms
crypto step	1 per 10 frames

policy	security level	total time	energy	deadline misses	early-exit rate
naive	768 fixed	78.4 s	3382 mJ	70	0%
static	768 fixed	69.6 s	3206 mJ	70	45%
adaptive	768 fixed	67.9 s	3173 mJ	70	56%
learned (joint)	**512 under tight slot**	**45.9 s**	**2079 mJ**	**0**	**85%**

The learned joint policy cuts total time by **41%** and energy by **39%** against naive, and removes **all 70 deadline misses**, by dropping to ML-KEM-512 when the deadline is tight and exiting early more often. Note the honest limitation: this is a simulator over measured costs, not a deployed controller, and the costs used are the bare-metal PQClean figures — the later -O2 wolfSSL benchmark measured the same 768 handshake 5.4x faster, which would change the numbers. Re-running against those is outstanding.

8. The classifiers, and what they actually show

Two leakage studies trained classifiers on device power traces. Both are reported with the caveat that emerged when they were re-audited.

study	target	classifier result
Activity leakage	which ML-KEM operation is running (keypair / encaps / decaps)	99.8% CV accuracy from a 256-point envelope
Exit leakage	which exit the network took (exit1 / 2 / 3)	98.3% full envelope, 100% with edges removed

The re-audit changes the interpretation. The three ML-KEM operations have non-overlapping execution times — keypair max 51.4 ms, encapsulate 54.3–57.7 ms, decapsulate 66.7–210.9 ms — with **zero cross-class overlap across 1,098 samples**. A duration-only classifier (one scalar, no power trace at all) scores **1.000**. The exit-leakage study independently lists a duration-only baseline at 100%.

So both results are reproducible, but what they demonstrate is a **timing** leak, not a power-envelope leak. The learned classifier adds nothing over a stopwatch. That is a weaker claim than the original framing and it is now stated as such. The consequence for the defence is direct: randomising the power envelope while the operation still takes different amounts of time hides nothing, so the countermeasure has to equalise duration.

9. Summary

Result	Value	Status
Exit accuracies	0.733 / 0.923 / 0.931 (baseline 0.945)	measured
Cost of anytime property	−1.4 points on the final exit	measured
Certified tiers	**6 of 9** pass the frozen protocol	certified
Learned policy, Bonferroni	exit1 1.47% certifies; exit2 3.61% does not	certified / not

ACI under shift	0.486 → 0.939 coverage at +3.3% compute	measured
LTC	NO-GO — frontier saturates at exit2 (99.0% of exit3 at 53% cost)	decision
Joint scheduler	−41% time, −39% energy, 0 misses vs naive	simulated
Leakage	99.8% / 98.3%, but a duration-only classifier scores 100%	re-audited